



Processors

As far as processors go, this was an out and out Intel year, which had its Sandy Bridge E series of processors at the beginning of the year and the Ivy Bridge processors launch in April. The flagship Ivy Bridge launch was followed by a wide range of SKUs for various budgets. This made the Sandy Bridge processor prices quite attractive. On the AMD front, last year we saw the 8-core FX-8150 which was taken over by the 4-core Intel Sandy Bridge Core i7-2600K in terms of performance, offered hardly any competition in that department. AMD did come out with its Trinity line of APUs towards the end of the year, which gave a brilliant integrated graphics performance as compared to an already impressive Llano APUs which were launched last year. We did not get the latest Piledriver processor – the FX-8350 from AMD at the time of writing, so its review can be expected next year.

Zero1 Winner

Intel Core i7-3960X

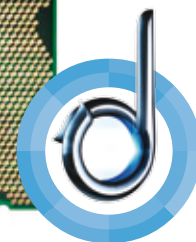
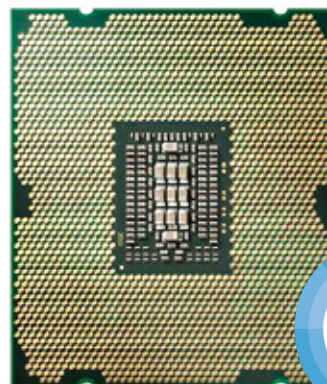
The Sandy Bridge E processor – the Core i7-3960X running at a clock speed of 3.3 GHz with a max Turbo frequency of 3.9GHz, with a shared L3 cache of 15MB, looks great on paper. There are lot of common aspects from the Sandy Bridge microarchitecture that have been carried forward. Performance scores were insane. Cinebench R11.5 gave a score of 11.27 as compared with 5.96 of the AMD FX-8150 and 6.79 of Core i7-2600K. X264 HD encoding at 720p gave a whopping 58.47FPS (FX-8150 @36.76). WinRAR compression/decompression happened at 4282 KB/s up from 3505 KB/s of the 2600K. If you are a power user and multi-threaded applications are what you indulge in the most, this is the beast of a processor for you. Of course, be ready to shell out a fortune for this processor.

Recommended Motherboard: ASUS Rampage IV Extreme

Worthy Mentions

Intel Core i7-3770K

Intel Sandy Bridge processors have been the favourite of mainstream system builders, but their integrated graphics namely HD 3000 and HD 2000 did not offer much in terms of gaming. With the first Intel Ivy Bridge processor, namely Core i7-3770K, we see an improvement on the gaming front. The Core i7-3770K is based on a 22nm process incorporating tri-gate transistors popularly known as 3D



▲ INTEL CORE I7-3960X

Price: ₹58,000

transistors, which allows faster switching of states, and packing in more transistors.

Intel HD 4000 graphics gave very good frame rates in most of the games. But the AMD A8-3850's HD 6550D graphics still gave superior frame rates. For Sandy Bridge users, it does not make much sense to upgrade to the i7-3770K.

For a pre-Sandy Bridge system builder this is a very good upgrade option if you are looking to build a high-end gaming and multimedia system.

Recommended Motherboard: ASUS Maximus V Extreme



Intel Core i7-3770K

Price: ₹20,000



AMD A10-5800K

Price: ₹7,000

AMD A10-5800K

AMD had us all impressed with the graphics performance of their Fusion series Llano APU – AMD A8-3850 last year. Carrying that tradition forward, AMD's latest APU - the A10-5800K, gives even better graphics performance. The quad-core A10-5800K is the flagship APU codenamed Trinity, has a base clock of 3.8 GHz which can scale up to 4.2 GHz on Turbo. It houses the 384-core Radeon HD 7660D GPU clocked at 800 MHz. This is quite an upgrade from the HD 6550D that was seen on the A8-3850.

The compute performance level is at par with the Intel Core i3 Ivy bridge processors. But the real strength of the A10-5800K lies in its gaming performance. *Resident Evil 5* (1680x1050, medium, DX10) gave 52 FPS; *STALKER: COP* (1680x1050, medium, DX11) gave a decent 32.4 FPS and *DIRT 3* (1680x1050, med) gave 42.1 fps. These readings were much better than HD 4000 on the Intel Core i7-3770K. If you are looking for a GPU-independent system, this is the processor for you. It offers the best gaming performance that we have seen on any processor, but is not so great at computing tasks.

Recommended Motherboard: Gigabyte GA F2A85X-UP4